7CS039: Coursework on Statistics For Data Science

Exploratory Data Analysis of Survival from Malignant Melanoma

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INTRODUCTION

This assignment uses the R programming language to present an exploratory data analysis (EDA) of a clinical dataset. With an emphasis on survival time and important prognostic factors, the analysis aims to summarize, visualize, and investigate relationships within the data. Without making strong modeling assumptions too early on, exploratory analysis is used to find patterns, variability, and possible associations. (r-project, 2025)

The Survival from Malignant Melanoma dataset, which includes clinical measurements from patients treated surgically for malignant melanoma in Denmark between 1962 and 1977, was examined. Survival time, patient status, age, sex, year of surgery, tumor thickness, and ulceration are among the variables. These factors offer a foundation for examining the relationship between patient demographics and tumor characteristics and survival outcomes.

DATA DESCRIPTION AND PREPARATION

Each of the 205 observations in the dataset represents a patient who received surgical treatment for malignant melanoma in Denmark between 1962 and 1977. Survival time (measured in days), patient status at the conclusion of follow-up, age, sex, year of surgery, tumor thickness (measured in millimeters), and the existence or lack of ulceration are among the variables.

To guarantee accurate interpretation in R, categorical variables were transformed into factors prior to analysis. The dataset was examined to make sure it was properly formatted and appropriate for exploratory analysis (see Figures 1).

```
> str(melanoma)
'data.frame':   205 obs. of  8 variables:
 $ X        : int  1 2 3 4 5 6 7 8 9 10 ...
 $ time     : int  10 30 35 99 185 204 210 232 232 279 ...
 $ status   : Factor w/ 3 levels "Died of melanoma",..: 3 3 2 3 1 1 1 3 1 1 ...
 $ sex      : Factor w/ 2 levels "Female","Male": 2 2 2 1 2 2 2 1 2 1 ...
 $ age      : int  76 56 41 71 52 28 77 60 49 68 ...
 $ year     : int  1972 1968 1977 1968 1965 1971 1972 1974 1968 1971 ...
 $ thickness: num  6.76 0.65 1.34 2.9 12.08 ...
 $ ulcer    : Factor w/ 2 levels "Absent","Present": 2 1 1 1 2 2 2 2 2 2 ...
```

```
> melanoma <- read.csv("/Users/oeze/Documents/wlv/7CS039/melanoma.csv")
> melanoma$sex <- factor(melanoma$sex, levels = c(0,1),
+                        labels = c("Female","Male"))
> melanoma$ulcer <- factor(melanoma$ulcer, levels = c(0,1),
+                          labels = c("Absent","Present"))
> melanoma$status <- factor(melanoma$status,
+                           levels = c(1,2,3),
+                           labels = c("Died of melanoma",
+                                      "Alive",
+                                      "Died of other causes"))
```

Figure 1: Structure of the malignant melanoma dataset as returned by str(melanoma) *and converted variables to appropriate formats*

I. SUMMARY STATISTICS OF THE DATASET

```
summary(melanoma$time)
summary(melanoma$age)
summary(melanoma$thickness)
```

```
table(melanoma$sex)
prop.table(table(melanoma$sex))

table(melanoma$ulcer)
table(melanoma$status)
```

Figure 2: R code for summary statistics for each of the variables in the melanoma dataset

The output after running the above R codes for statistical summary for the dataset are been described below:

Of the 205 patients in the sample, 126 (61.5%) are female and 79 (38.5%) are male, suggesting a greater proportion of female patients. When analyzing any gender-based comparisons, this discrepancy should be taken into account.

Patients' survival times range widely, from 10 days to 5,565 days (15.3 years). With a mean survival duration of 2,153 days and a median survival time of 2,005 days (about 5.5 years), the distribution is right-skewed and driven by long-term survivors.

By the time the follow-up period ended, 134 patients (65.4%) were still living, 57 patients (27.8%) had passed away from melanoma, and 14 patients (6.8%) had died from other causes. This suggests that melanoma-specific mortality is still significant even if the majority of patients survived the study period.

Patients vary in age from 4 to 95 at the time of surgery, with a mean age of 52.46 and a median age of 54. The majority of patients were middle-aged to older people, with a little bias towards older ages, according to the interquartile range (42 to 65 years).

The thickness of the tumor varies significantly, from 0.10 mm to 17.42 mm. The mean thickness is 2.92 mm, but the median thickness is 1.94 mm, indicating a significant right skewness brought on by a few extremely large tumors.

The fact that ulceration is present in 90 patients (43.9%) and absent in 115 patients (56.1%) suggests that ulceration is somewhat widespread in this cohort and is thus a crucial variable for more research.

```
> summary(melanoma)
       X            time                        status        sex           age             year
 Min.   :  1   Min.   :  10   Died of melanoma    : 57   Female:126   Min.   : 4.00   Min.   :1962
 1st Qu.: 52   1st Qu.:1525   Alive               :134   Male  : 79   1st Qu.:42.00   1st Qu.:1968
 Median :103   Median :2005   Died of other causes: 14                Median :54.00   Median :1970
 Mean   :103   Mean   :2153                                           Mean   :52.46   Mean   :1970
 3rd Qu.:154   3rd Qu.:3042                                           3rd Qu.:65.00   3rd Qu.:1972
 Max.   :205   Max.   :5565                                           Max.   :95.00   Max.   :1977
   thickness         ulcer
 Min.   : 0.10   Absent :115
 1st Qu.: 0.97   Present: 90
 Median : 1.94
 Mean   : 2.92
 3rd Qu.: 3.56
 Max.   :17.42
```

```
> summary(melanoma$time)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
     10    1525    2005    2153    3042    5565
> summary(melanoma$age)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   4.00   42.00   54.00   52.46   65.00   95.00
> summary(melanoma$thickness)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.10    0.97    1.94    2.92    3.56   17.42
> table(melanoma$sex)

Female   Male
   126     79
> prop.table(table(melanoma$sex))

   Female      Male
0.6146341 0.3853659
> table(melanoma$ulcer)
```

Figure 2.1: Summary statistics output for the melanoma dataset generated using summary(melanoma)

II. GRAPHICAL SUMMARIES AND EMERGING TRENDS

```
# Histograms
hist(melanoma$time,
     main="Distribution of Survival Time",
     xlab="Survival Time (days)")
```

```
hist(melanoma$age,
     main="Distribution of Age",
     xlab="Age (years)")
```

```
hist(melanoma$thickness,
     main="Distribution of Tumour Thickness",
     xlab="Thickness (mm)")
```

```
# Boxplots
boxplot(time ~ sex, data=melanoma,
        main="Survival Time by Gender",
        xlab="Gender",
        ylab="Survival Time (days)")
```

```
boxplot(thickness ~ ulcer, data=melanoma,
        main="Tumour Thickness by Ulceration Status",
        xlab="Ulceration",
        ylab="Thickness (mm)")
```

```
# Bar plot
barplot(table(melanoma$status),
        main="Patient Status at End of Study"
        ylab="Frequency")
```

Figure 3: R code to geenrate a graphical summary for the dataset

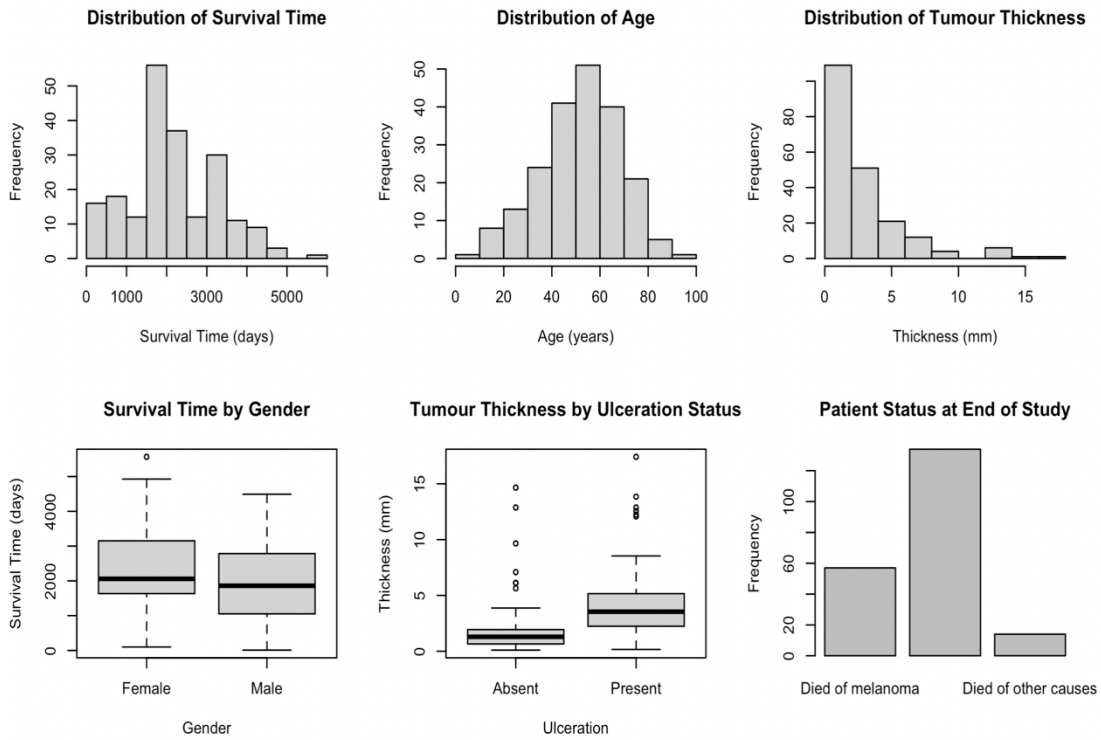

Figure 3.1: Graphical summaries of the melanoma dataset, including distributions of survival time, age, and tumour thickness, and comparisons by gender, ulceration status, and patient outcome.

Based on the above graph that summaries the melanoma dataset distribution the following were deduced. (Devashree, 2024)

Using the R code, I was able to create a high-quality, graphs with bar charts, boxplot and histograms, which facilitate effective data analysis and presentation.

The distribution of survival time is highly skewed to the right, with the majority of patients living for a few thousand days and a tiny percentage living for considerably longer. This suggests a great deal of variation in the results after surgery.

The majority of patients fall between their early 40s and mid-60s, with an age distribution that is roughly typical and centered around the mid-50s. Extreme ages are not very common.

The majority of tumors are thin, with a few extremely thick tumors serving as outliers. The thickness of tumors is strongly skewed to the right. This implies significant variation in the severity of the illness upon diagnosis.

Although there is significant overlap between the two groups, females have a little longer median survival time than men, suggesting that survival is not significantly influenced by gender alone.

Because ulceration is linked to more advanced illness, tumors with ulceration are often thicker and more varied than non-ulcerated tumors.

The majority of patients were still alive at the conclusion of the trial, with relatively few fatalities from other causes and fewer deaths linked to melanoma, suggesting that although melanoma-specific mortality was important, it only impacted a small percentage of patients.

III. REGRESSION AND CORRELATION ANALYSIS

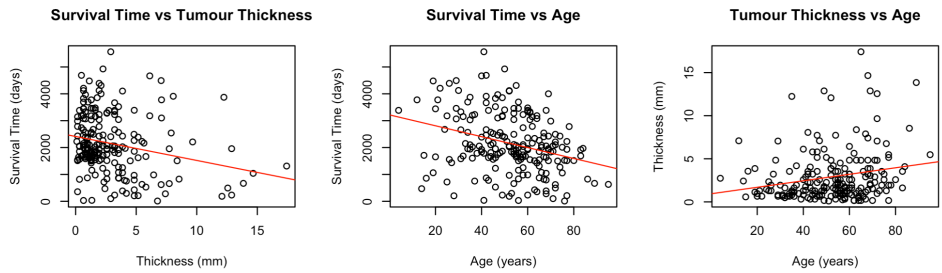

Figure 4: Scatterplots with fitted linear regression lines illustrating the relationships between survival time and tumour thickness, survival time and age, and tumour thickness and age.

```
# Scatterplots with regression lines
plot(time ~ thickness, data=melanoma,
     main="Survival Time vs Tumour Thickness",
     xlab="Thickness (mm)",
     ylab="Survival Time (days)")
abline(lm(time ~ thickness, data=melanoma), col="red")

plot(time ~ age, data=melanoma,
     main="Survival Time vs Age",
     xlab="Age (years)",
     ylab="Survival Time (days)")
abline(lm(time ~ age, data=melanoma), col="red")

plot(thickness ~ age, data=melanoma,
     main="Tumour Thickness vs Age",
     xlab="Age (years)",
     ylab="Thickness (mm)")
abline(lm(thickness ~ age, data=melanoma), col="red")

# Linear regression models
model_time_thickness <- lm(time ~ thickness, data=melanoma)
summary(model_time_thickness)

model_time_age <- lm(time ~ age, data=melanoma)
summary(model_time_age)

model_thickness_age <- lm(thickness ~ age, data=melanoma)
summary(model_thickness_age)

# Correlations
cor(melanoma$time, melanoma$thickness)
cor(melanoma$time, melanoma$age)
cor(melanoma$thickness, melanoma$age)
```

Figure 4.1: R code for the Scatterplots with fitted linear regression lines and regression.

The associations between survival time, tumor thickness, and age were investigated using linear regression. Figure 4 displays scatterplots with fitted regression lines.

Tumor thickness and survival duration have a statistically significant inverse connection. According to the model, the survival duration decreases by around 89 days for every millimeter that the tumor thickness increases ($p < 0.001$). Nevertheless, the low explained variance ($R2 = 0.055$) indicates that thickness by itself is a poor indicator of survival. The slight negative correlation ($r = -0.235$) is in line with this.

Age and survival time are also found to be significantly negatively correlated. With a somewhat better but still low explanatory power ($R2 = 0.091$), survival time reduces by around 20 days for every extra year of age ($p < 0.001$). A moderately negative linear connection is shown by the correlation coefficient ($r = -0.302$).

According to the regression model, tumor thickness rises by around 0.038 mm year ($p = 0.002$). A slight positive correlation ($r = 0.212$) supports this link, which explains very little variance in thickness ($R^2 = 0.045$).

In general, older individuals often have somewhat bigger tumors, but thicker tumors are linked to shorter survival times. All of the models' poor R2 values suggest that survival outcomes are impacted by a variety of factors outside the scope of this study.

IV. COMMENTARY ON OBSERVED RELATIONSHIPS

According to the analysis, survival time is adversely correlated with both tumor thickness and age, with greater age and bigger tumors being associated with shorter survival. These correlations show significant unexplained variability in outcomes and are statistically significant but weak.

There is a small positive correlation between age and tumor thickness, but a somewhat greater correlation between age and survival time. Overall, the findings imply that while tumor thickness and age have an impact on survival, they are insufficient to fully account for patient outcomes.

V. TWO-SAMPLE SIGNIFICANCE TESTS BY GENDER

```
t.test(time ~ sex, data=melanoma)
t.test(age ~ sex, data=melanoma)
wilcox.test(thickness ~ sex, data=melanoma)
```

Figure 5: R code for a sample significance tests by gender using Welch two-sample t-test and Wilcoxon rank-sum test

The survival times of males and females were compared using a Welch two-sample t-test. The findings show that there is a statistically significant difference in the mean survival time between the sexes (p = 0.039), with females having a greater mean survival period (2,283 days) than men (1,946 days). This implies that after surgery, female patients often have greater survival rates.

There is no statistically significant difference between males and females when age is compared by gender using a Welch two-sample t-test (p = 0.341). Given that the mean ages of male and female patients are comparable, it is doubtful that aging would obscure the gender disparities in survival that have been noted.

Because of its skewed distribution, a Wilcoxon rank-sum test was utilized to compare tumor thickness by gender. Tumor thickness distributions vary by gender, according to the test, which shows a statistically significant variation in tumor thickness between males and girls (p = 0.004).

This test is used to compare two independent groups in order to determine whether there is a statistically significant difference between their means (Curran, 2013).

```
> wilcox.test(thickness ~ sex, data=melanoma)

        Wilcoxon rank sum test with continuity correction

data:  thickness by sex
W = 3794.5, p-value = 0.004213
alternative hypothesis: true location shift is not equal t
o 0
```

```
> t.test(age ~ sex, data=melanoma)

        Welch Two Sample t-test

data:  age by sex
t = -0.95559, df = 154.42, p-value = 0.3408
alternative hypothesis: true difference in means between g
roup Female and group Male is not equal to 0
95 percent confidence interval:
 -7.162764  2.492280
sample estimates:
mean in group Female   mean in group Male
            51.56349             53.89873
```

```
> t.test(time ~ sex, data=melanoma)

        Welch Two Sample t-test

data:  time by sex
t = 2.0848, df = 159.27, p-value = 0.03868
alternative hypothesis: true difference in means between g
roup Female and group Male is not equal to 0
95 percent confidence interval:
  17.74767 656.12032
sample estimates:
mean in group Female   mean in group Male
            2282.643             1945.709
```

Figure 5.1: Logs from test using Welch two-sample t-test and Wilcoxon rank-sum test

VI. DISTRIBUTIONAL ASSESSMENT USING QQ-PLOTS

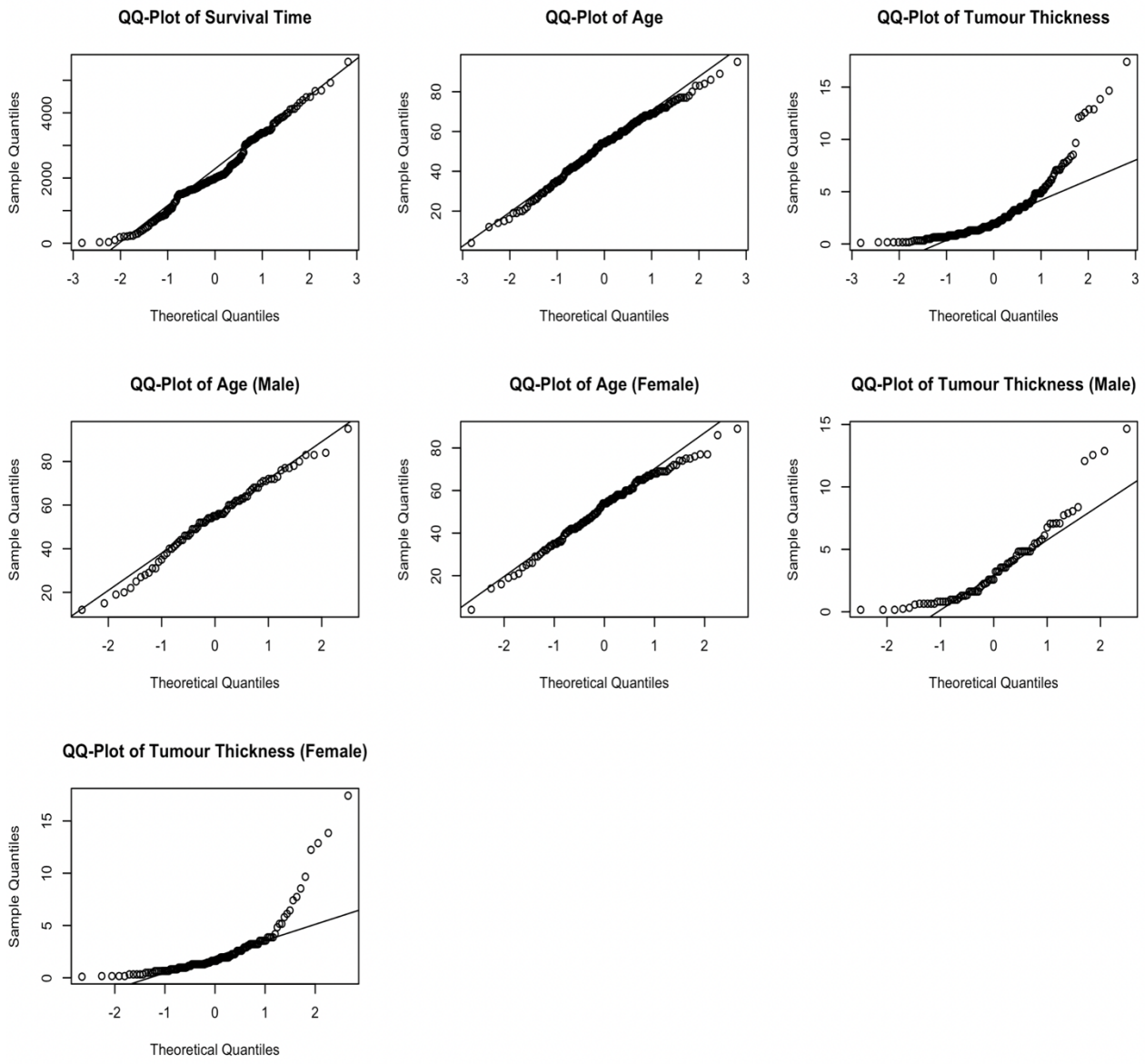

Figure 6: Normal Q–Q plots illustrating the distributional properties of survival time, age, and tumour thickness, including gender-specific comparisons**.**

Extreme survival values are seen in the QQ-plot of survival time, which clearly exhibits right skewness and a significant departure from normalcy in the top tail. The large variability shown in the regression graphs is consistent with this non-normal (Field, 2012) distribution.

Age is roughly normally distributed with just little tail deviations, according to the QQ-plots of age by gender and overall, which nearly follow the reference line. Comparable age distributions across genders are suggested by the similarities between the male and female charts.

Tumor thickness QQ-plots show significant deviations from normalcy, with severe top values and a noticeable right skewness. Both boys and females exhibit this pattern, which validates the application of non-parametric techniques for comparisons pertaining to thickness.

For survival time vs tumor thickness and survival time versus age, the regression lines exhibit negative trends, suggesting decreased survival with increasing thickness and age, respectively. Although all correlations indicate significant heterogeneity, there is a small positive trend between tumor thickness and age, suggesting somewhat thicker tumors among older individuals. (Ott, 2015)

```r
qqnorm(melanoma$time,
       main = "QQ-Plot of Survival Time")
qqline(melanoma$time)

qqnorm(melanoma$age,
       main = "QQ-Plot of Age")
qqline(melanoma$age)
```

```r
qqnorm(melanoma$thickness,
       main = "QQ-Plot of Tumour Thickness")
qqline(melanoma$thickness)

qqnorm(melanoma$age[melanoma$sex == "Male"],
       main = "QQ-Plot of Age (Male)")
qqline(melanoma$age[melanoma$sex == "Male"])
```

```r
qqnorm(melanoma$age[melanoma$sex == "Female"],
       main = "QQ-Plot of Age (Female)")
qqline(melanoma$age[melanoma$sex == "Female"])

qqnorm(melanoma$thickness[melanoma$sex == "Male"],
       main = "QQ-Plot of Tumour Thickness (Male)")
qqline(melanoma$thickness[melanoma$sex == "Male"])
```

```r
qqnorm(melanoma$thickness[melanoma$sex == "Female"],
       main = "QQ-Plot of Tumour Thickness (Female)")
qqline(melanoma$thickness[melanoma$sex == "Female"])
```

Figure 6.1: R Code to generate normal Q–Q plots to comparisons the distribution on survival time, age, and tumour thickness, including gender-specific

VII. INSIGHTS, DISCUSSION AND RECOMMENDATIONS

I discovered the following remarkable insights from the thorough exploratory examination of the provided dataset:

i. Existance of Temporal Trends: The majority of operations took place between 1971 and 1974, with notable exceptions in 1963, 1975, and 1976. Changes in healthcare access, diagnostic procedures, or surgical capabilities over the research period may be reflected in this temporal grouping. Outliers in later years (1977) are shown by the boxplot analysis, indicating that some patients had surgery beyond the initial data collecting period, which may have an impact on representativeness. (Erdmann, 2013)

ii. Factors Affecting Survival Time: One noteworthy conclusion is that very few individuals died from unrelated reasons, a significant percentage of patients survived, and a small number of patients died from melanoma. This suggests that other variables, such comorbidities or the success of surgical surgery, could have affected the results. Ulceration may be a predictive factor due to its high prevalence (56% of patients). If there is a strong correlation between ulceration and worse survival results, this might be clarified by further investigation.

iii. Variable Relationships and Normality: The low R2 value in the regression indicates that the link is not strong enough to explain most of the variance, but the weak negative correlation between time and tumor thickness shows that survival time gradually reduces as tumor thickness grows.
Similarly, older patients may have lower survival durations after surgery, although the impact magnitude is small, according to the weak negative association between time and age.

Following a thorough examination of the provided data, I discovered the following recommendations that might result in a more precise and effective statistical approach to the data:

i. According to the data, there are significant variations in patient outcomes, with survival time among melanoma patients being extremely variable and right-skewed. Although its overall explanatory power is modest, tumor thickness is regularly linked to decreased survival, indicating its significance as a prognostic indicator.

ii. Additionally, there is a negative correlation between age and survival time; older patients often have worse results. Age and tumor thickness have a weakly positive correlation, indicating that disease severity may modestly rise with age. Despite similar age distributions, gender-based studies show that female patients had higher survival times and that there are notable disparities in tumor thickness.

iii. Overall, the results indicate that a variety of variables affect melanoma survival, with tumor thickness, age, and gender having a minor impact. To further explain variations in patient prognosis, future research should concentrate on multivariable survival models and look at other clinical variables including ulceration. (Jemal, 2011)

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